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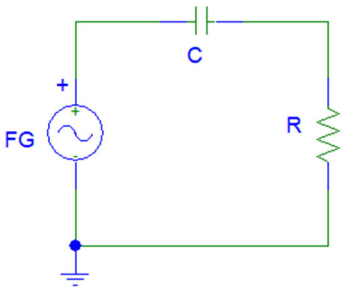
Aim

To be able to practically perform relevant measurements and analyze the measurement results.
Get an understanding for characteristic properties for an alternating current.
To verify characteristics for OP AMP.

Task 1

Measure how the phase shift and the impedance change when the frequency is changing.

Connection diagram:



$R = 330\Omega$
$C = 1\mu F$
Voltmeter TENMA
Oscilloscope HM407
FG = Function Generator

Figure 1. Connection diagram for Task 1.

Procedure:

Connect according to the Connection diagram in figure 1. Use a Breadboard.

We will take the amplitude A and frequency values f from the oscilloscope image. The frequency is determined by measuring the period T and then calculating the frequency using the relationship $f \cdot T = 1$.
We will need to use this method throughout the lab to determine the frequency.

In a) and b) we use the oscilloscope (Channel 1) to measure the output signal from the function generator.

In c) and d) we use the oscilloscope (Channel 1) to measure the voltage over R and the Channel 2 to measure the output signal from the function generator.

The oscilloscope must be AC-coupled.

a)

Output signal from function generator:
Sinusoidal with amplitude $2.0V$, offset $0V$, and frequency $200Hz$, see figure 2.

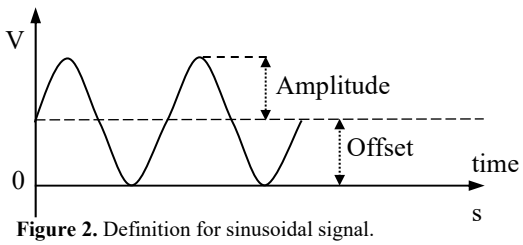


Figure 2. Definition for sinusoidal signal.

Measure the voltage over R and the voltage over C **with voltmeter** (use two voltmeters).
Note that the voltage over R is a measure of the current. The value of the current you get by means of Ohm's law.

(Task 1, continues next page)

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(Task 1, continues)

b)

Output signal from function generator:
Sinusoidal with amplitude $2.0V$, offset $0V$, and frequency $1kHz$, see figure 2.

Measure the voltage over R and the voltage over C **with voltmeter**.

c)

Output signal from function generator:
Sinusoidal with amplitude $2.0V$, offset $0V$, and frequency $200Hz$, see figure 2.

Measure the phase shift between the voltage and the current in the circuit **with Oscilloscope**. Channel 1 to measure the voltage over R and the Channel 2 to measure the output signal from the function generator.

d)

Output signal from function generator:
Sinusoidal with amplitude $2.0V$, offset $0V$, and frequency $1kHz$, see figure 2.

Measure the phase shift between the voltage and the current in the circuit **with Oscilloscope**.

What to show the Lab Instructor during the lab:

a) and **b)**

Images from oscilloscope and scales setting.

Measured values V_R and V_C.

c) and **d)**

Images from oscilloscope and scales setting.

Measured values on the phase shift.

What to include in the lab report:

a) and **b)**

Images from oscilloscope and scales setting.
Measured values.
Theoretical value $X_C = \frac{1}{2\pi f C}$.
Compare the measured values with the theoretical values.
Conclusions.

c) and **d)**

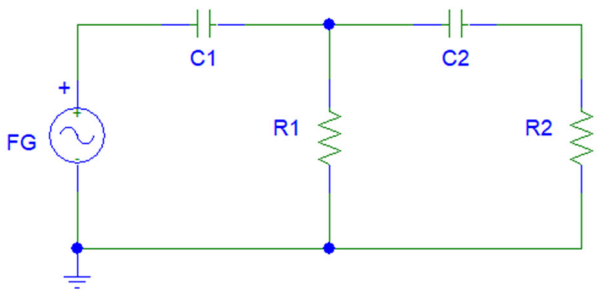
Images from oscilloscope and scales setting.
Measured values.
Theoretical value $\varphi = \arctan\frac{X_C}{R}$
Compare the measured values with the theoretical values.
Conclusions.

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Task 2

To measure a voltage and a current and compare by means of complex method with the theoretical values.

Connection diagram:



input = Function Generator
$R_1 = 1k\Omega$
$R_2 = 2.2k\Omega$
$C_1 = 0.47\mu F$
$C_2 = 0.1\mu F$

Figure 3. Connection diagram for Task 2.

Procedure:

Connect according to the Connection diagram in figure 3. Use a Breadboard.

We use the oscilloscope (Channel 1) to measure the voltage over R_2 and the Channel 2 to measure the output signal from the function generator.

The oscilloscope must be AC-coupled.

Output signal from function generator:
Sinusoidal with amplitude $6.0V$, offset $0V$, and frequency $1kHz$, see figure 2.

Measure the voltage $\hat{v}_{R_2}$ and the output signal amplitude. Measure also the phase shift between the signals.

What to show the Lab Instructor during the lab:

How you have built up the circuit.

Images from oscilloscope and scales setting.

What to include in the lab report:

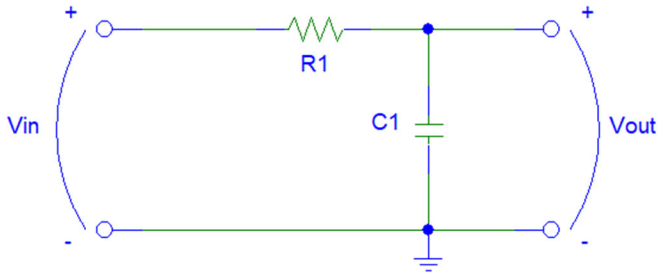
Images from oscilloscope and scales setting.
Measured values.
Theoretical value on the current $\underline{I}_{R_2}$. Use input signal as reference in the calculation.
Compare the measured values with the theoretical values.
Conclusions.

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Task 3

Measure the magnitude function $A_v(f)=\left|\frac{V_{out}}{V_{in}}\right|$ and the phase function $\varphi(f)=\arg\left(\frac{V_{out}}{V_{in}}\right)$ on RC-filter.

Connection diagram:



input = Function Generator
$R_1=1k\Omega$
$C_1=0.1\mu F$

Figure 4. Connection diagram for Task 3.

Procedure:

Connect according to the Connection diagram in figure 4. Use a Breadboard.

We use the Channel 1 to measure the voltage V_{in} and the Channel 2 to measure the voltage V_{out}.

Output signal from function generator:
Sinusoidal with amplitude $3.0V$, offset $0V$, and frequency within the frequency range $100Hz-20kHz$, see figure 2.

Measure the $A_v(f)=\left|\frac{V_{out}}{V_{in}}\right|$ and $\varphi(f)=\arg\left(\frac{V_{out}}{V_{in}}\right)$ within the frequency range $100Hz-20kHz$. Make 5 measurements *under* cut-off frequency and 5 measurements *over* cut-off frequency, *include* the cutoff frequency in your measurement. We use V_{in} as the reference when measuring the phase.

What to show the Lab Instructor during the lab:

How you have built up the circuit.

Measured values $A_v=\left|\frac{V_{out}}{V_{in}}\right|$ and $\varphi=\arg\left(\frac{V_{out}}{V_{in}}\right)$.

What to include in the lab report:

Diagram for $\left|\frac{V_{out}}{V_{in}}\right|=f(f)$ and $\arg\left(\frac{V_{out}}{V_{in}}\right)=f(f)$. Measured and theoretical values in the same diagram.

Theoretical expression for A_v is $A_v(\omega)=\frac{1}{1+j\omega R_1C_1}\underset{\omega_1=\frac{1}{R_1C_1}}{=}\frac{1}{1+j\frac{\omega}{\omega_1}}$, where ω_1 is the cutoff frequency.

A_v can also be express as $A_v(f)=\frac{1}{1+j2\pi fR_1C_1}\underset{f_1=\frac{1}{2\pi R_1C_1}}{=}\frac{1}{1+j\frac{f}{f_1}}$, where f_1 is the cutoff frequency.

Compare the measured values with the theoretical values.
Conclusions.

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Task 4

See how not infinite gain and not infinitive bandwidth affects the output from an OP AMP.

Theory:

The gain for an OP AMP changes with the frequency, see figure 5. For the OP AMP 741 the open loop gain at dc is $A_0 = 200000$ and the cut off frequency is $f_C = 5Hz$.

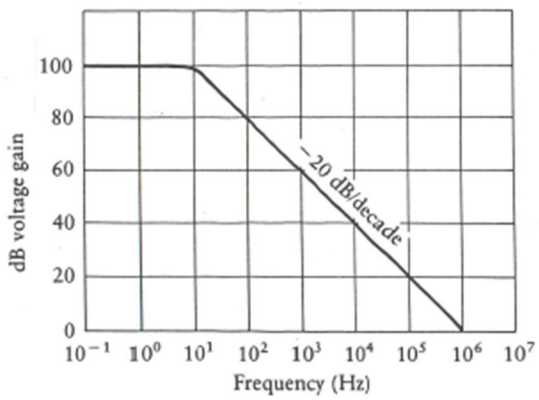


Figure 5. Open loop gain response for OP AMP 741 with logarithmic axes.

Mathematically can we describe the curve in figure 5 with $A(f) = \dfrac{A_0}{1 + j\dfrac{f}{f_C}}$.

Gain-**B**andwidth-Product GB is an important parameter for OP AMPs. For the open loop gain in figure 5 is $GB = A_0 \cdot f_C = 200000 \cdot 5 = 1000000 = 1 \cdot 10^6\, Hz$. This value is constant on the curve in figure 5.

For a non-inverting OP AMP circuit, se figure 7, we have $\dfrac{v_{out}}{v_{in}} = \dfrac{R_1 + R_2}{R_1}$ where $\dfrac{R_1 + R_2}{R_1}$ is the amplification factor. The invers value $\dfrac{R_1}{R_1 + R_2}$ is called feedback factor and labeled with $\beta = \dfrac{R_1}{R_1 + R_2}$.

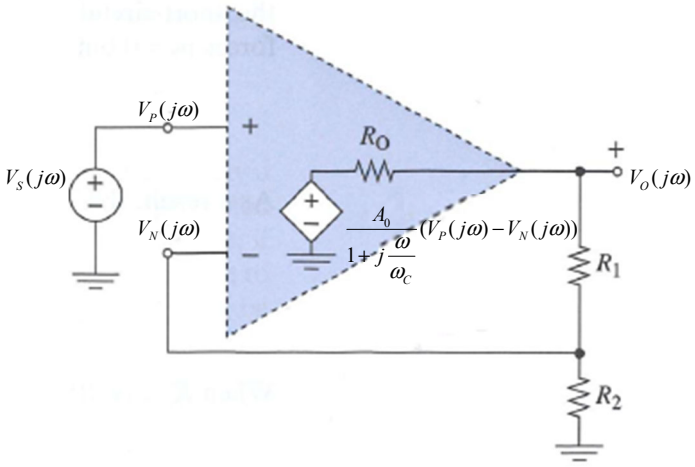


Figure 6. OP AMP model, not with infinite gain and not with infinitive bandwidth.

(Task 4, continues next page)

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(Task 4, continues)

If we use the OP AMP model in figure 6 with complex notification and $R_O = 0\Omega$ we have the following three equations:

$$\begin{cases} (1) \quad V_O(j\omega) = \dfrac{A_0}{1 + j\dfrac{\omega}{\omega_C}}(V_P(j\omega) - V_N(j\omega)) \\ (2) \quad V_P(j\omega) = V_S(j\omega) \\ (3) \quad V_N(j\omega) = V_O(j\omega)\dfrac{R_1}{R_1 + R_2} = V_O(j\omega)\beta \end{cases}$$

With these three equations we get: $\dfrac{V_O(j\omega)}{V_S(j\omega)} = \dfrac{A_0}{(1 + A_0\beta)(1 + j\dfrac{\omega}{\omega_C(1 + A_0\beta)})} =$

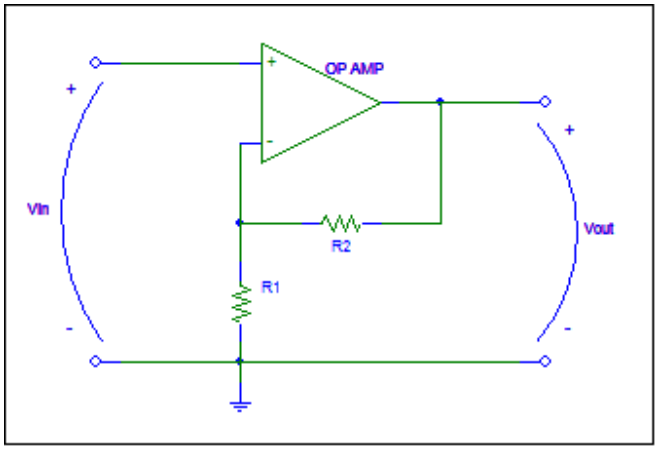
$$= \frac{A_0}{1 + A_0\beta}\frac{1}{1 + j\dfrac{\omega}{\omega_C(1 + A_0\beta)}} = \frac{A_0}{1 + A_0\beta}\frac{1}{1 + j\dfrac{f}{f_C(1 + A_0\beta)}}$$

The gain-bandwidth product for the closed-loop circuit is:

$\dfrac{A_0}{1 + A_0\beta} \cdot f_C(1 + A_0\beta) = \dfrac{A_0}{\cancel{1 + A_0\beta}} \cdot f_C(\cancel{1 + A_0\beta}) = A_0 \cdot f_C = 1 \cdot 10^6\, Hz$ which is the same as the open-loop case. In other words, the gain-bandwidth product is invariant and is not changed by feedback.

Connection diagram:

When writing connection diagram for an amplifier-constructions with OP AMPs, the power supply is usually not included in the schedule. We follow that principle in figure 7.



$V_{CC} = \pm 15.0V$
OP AMP 741
$R_1 = 1k\Omega$
$R_2 =$ decade resistor

Figure 7. Connection diagram for a non-inverting amplifier circuit, not including power supply.

(Task 4, continues next page)

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(Task 4, continues)

Procedure:

Follow the instructions in Lab.3 (Task 3 and Task 4) when we create power supply for the OP AMP and connect according to the Connection diagram.

The output cable from the function generator is red and black. Change the color for red cable when we go over to small cables. Choose orange or green color.

Connect according to the Connection diagram in figure 7. Use a Breadboard.

Turn on the power supply.

The oscilloscope must be AC-coupled.

a)

Set the decade resistor to $R_2 = 199k\Omega$.

Output signal from function generator:
Sinusoidal with amplitude $20mV$, offset $0V$, and frequency within the frequency range $100Hz - 10kHz$, see figure 2.

Use both channels of the oscilloscope, so that we can measure the input and output signals simultaneously. On channel 1 we measure input and on channel 2 we measure output.

Measure the input voltage $\hat{v}_{in}$ and the output voltage $\hat{v}_{out}$. Make 3 measurements *under* cut-off frequency and 3 measurements *over* cut-off frequency, *include* the cutoff frequency in your measurement.

b)

Set the decade resistor to $R_2 = 399k\Omega$.

Output signal from function generator:
Sinusoidal with amplitude $20mV$, offset $0V$, and frequency within the frequency range $100Hz - 10kHz$, see figure 2.

Measure the input voltage $\hat{v}_{in}$ and the output voltage $\hat{v}_{out}$. Make 3 measurements *under* cut-off frequency and 3 measurements *over* cut-off frequency, *include* the cutoff frequency in your measurement.

(Task 4, continues next page)

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(Task 4, continues)

What to show the Lab Instructor during the lab:

a) and **b)**

How you have built up the circuit.

Images from oscilloscope and scales setting.

Value on the upper cut-off frequency.

Value on the **G**ain-**B**andwidth-Product GB.

What to include in the lab report:

a) and **b)**

Images from oscilloscope and scales setting.
Value on the upper cut-off frequency.
Value on the **G**ain-**B**andwidth-Product GB.
Compare the measured values with the theoretical values.
Conclusions.

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Task 5

See how the Slew Rate affects the output from an OP AMP.

Theory:

Slew Rate is the maximal slope on the OP AMPs output. For OP AMP 741 a typical value is $0.5\frac{V}{\mu s}$. The unit shows that slew rate is equal to $\frac{\Delta V}{\Delta t}$.

The signal in this task is sinusoidal: $v(t) = \hat{v}\sin(\omega t)$, there $\omega = 2\pi f$.

So, to find the slew rate we derivative $v(t) \Rightarrow \frac{dv(t)}{dt} = \omega\hat{v}\cos(\omega t)$.

To find the max value of the frequency for the signal, we need to find when the slope is maximal. So, we derivative the slew rate $\frac{dv(t)}{dt} = \omega\hat{v}\cos(\omega t) \Rightarrow \frac{d^2v(t)}{dt2} = -\omega^2\hat{v}\sin(\omega t)$.

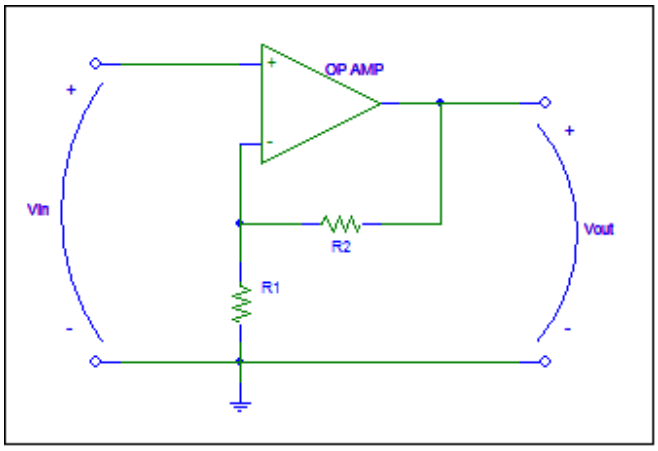
Maximal $\Rightarrow \frac{d^2v(t)}{dt2} = 0 \Rightarrow -\omega^2\hat{v}\sin(\omega t) = 0 \Rightarrow t = 0s$.

This gives us: slew rate $= \left.\frac{dv(t)}{dt}\right|_{\max} \underbrace{=}_{t=0s} \omega\hat{v}\cos(\omega\cdot 0) = \omega\hat{v} = 2\pi f_{\max}\hat{v} \Rightarrow$

$\Rightarrow f_{\max} = \frac{\text{slew rate}}{2\pi\hat{v}} = \frac{0.5\cdot 10^6}{2\pi\hat{v}} Hz$

Connection diagram:

When writing connection diagram for an amplifier-constructions with OP AMPs, the power supply is usually not included in the schedule. We follow that principle in figure 8.



$V_{CC} = \pm 15.0V$
OP AMP 741
$R_1 = 1k\Omega$
$R_2 = 2.2k\Omega$

Figure 8. Connection diagram for a non-inverting amplifier circuit, not including power supply.

(Task 5, continues next page)

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(Task 5, continues)

Procedure:

Follow the instructions in Lab.3 (Task 3 and Task 4) when we create power supply for the OP AMP and connect according to the Connection diagram.

The output cable from the function generator is red and black. Change the color for red cable when we go over to small cables. Choose orange or green color.

Connect according to the Connection diagram in figure 8. Use a Breadboard.

Turn on the power supply.

The oscilloscope must be AC-coupled.

a)

Output signal from function generator:

Sinusoidal with amplitude $3.0V$, offset $0V$, and frequency $\frac{f_{max}}{2}$, see figure 2.

Use both channels of the oscilloscope, so that we can measure the input and output signals simultaneously. On channel 1 we measure input and on channel 2 we measure output.

Measure (compare) the input voltage and the output voltage.

b)

Output signal from function generator:

Sinusoidal with amplitude $3.0V$, offset $0V$, and frequency $2 \cdot f_{max}$, see figure 2.

Measure (compare) the input voltage and the output voltage.

What to show the Lab Instructor during the lab:

a) and **b)**

How you have built up the circuit.

Images from oscilloscope and scales setting.

Value on f_{max}.

What to include in the lab report:

a) and **b)**

Images from oscilloscope and scales setting.
Value on f_{max}.
Theoretical calculation to find f_{max} if the signal is: $v(t) = \hat{v}\cos(\omega t)$, there $\omega = 2\pi f$.
Compare the measured values with the theoretical values.
Conclusions.